Gravitational Memory in Black Hole Mergers from Scattering Amplitudes

Jun 15, 2026,　Theoretical Developments in Gravitational Wave Physics

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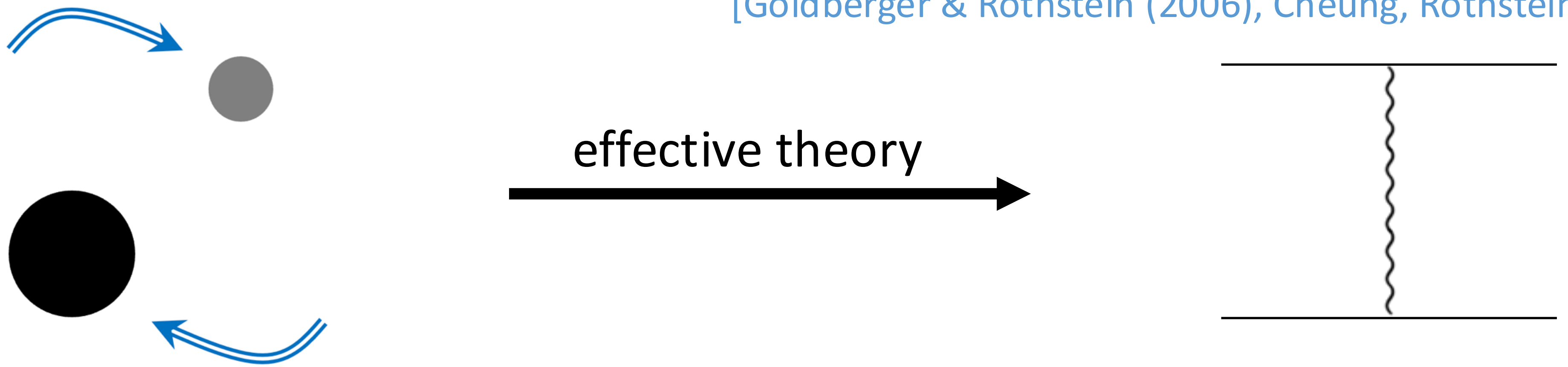
Work In Progress

Effective theory of Black Holes

The interaction in the 2-body system is captured by the graviton exchange.

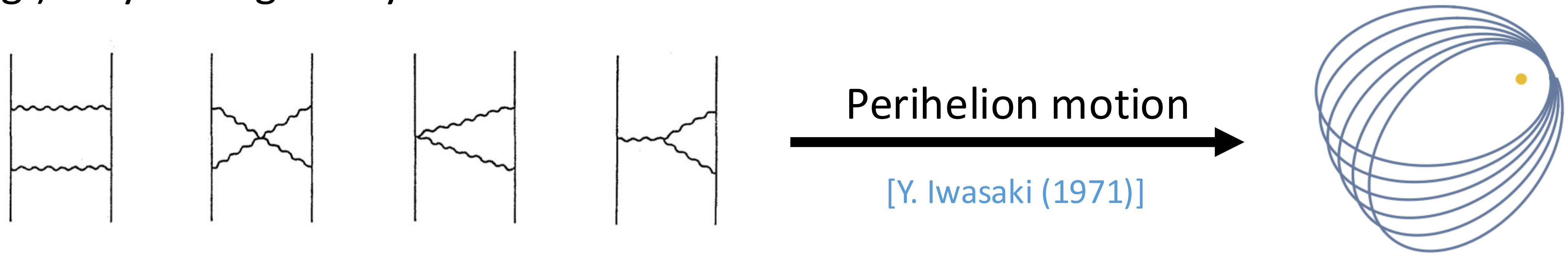
[Goldberger & Rothstein (2006), Cheung, Rothstein & Solon (2018)]



Finite-size effects (Love number *etc.*) enter only at high orders in G_N.

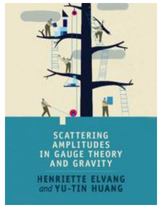
[Cheung & Solon (2020), Bern *et al.* (2018)]

e.g.) 50 years ago in Kyoto



On-Shell Program

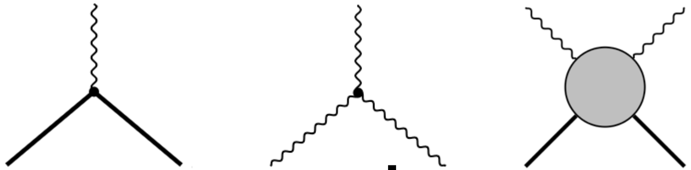
Reframing



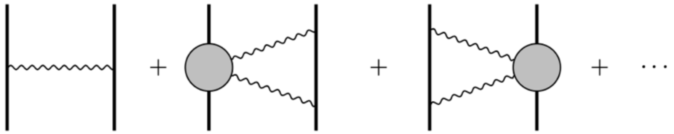
Attempts to determine scattering amplitudes without a Lagrangian → works well!

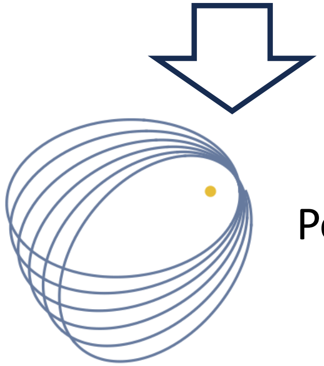
[N. Arkani-Hamed, T. C. Huang, Y-T. Huang (2018), …]

Poincaré symm. ＋ Locality



Unitarity





Perihelion shift

A lot of merits for the **graviton scattering**:

① No need for the gauge fixing

② PM expansion ≈ loop expansion [Z. Bern *et al.* (2018), …]

③ self force ≈ loop calculation [C. Cheung *et al.* (2023), …]

④ various methods : *double copy, generalized unitarity, IBP reduction, BCFW recursion etc. …*

→On-shell program!

How **classical** physics appears from **quantum** scattering amplitudes?

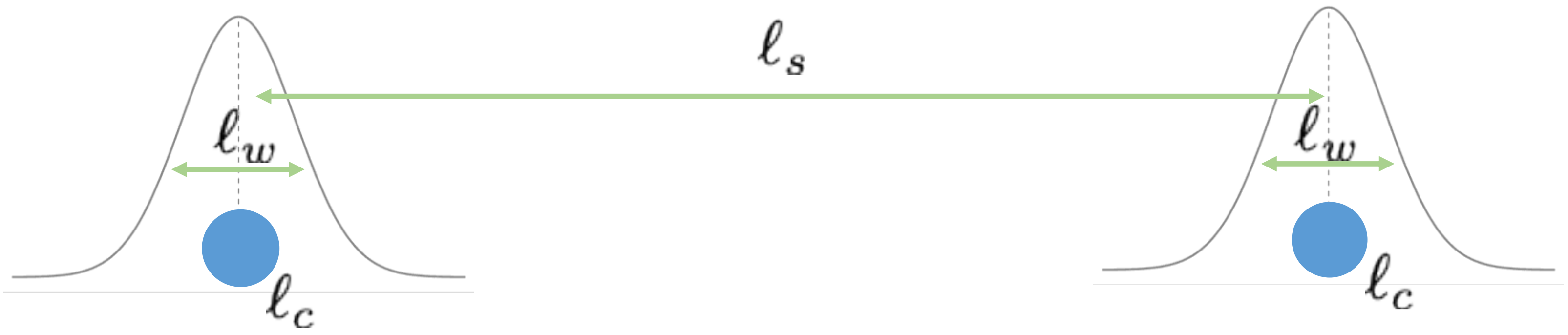
Setup1: KMOC formalism

[Kosower, Maybee, O'Connell (2019)]

wavepacket: $|\Psi\rangle = \int_{p_1,p_2} \phi(p_1)\phi(p_2)e^{i(b_1\cdot p_1 + b_2\cdot p_2)}\,|p_1;p_2\rangle$

On-shell integration measure:

$$\int_p := \int \frac{\mathrm{d}^4 p}{(2\pi)^3}\Theta(p^0)\delta(p^2 - M_p^2)$$



Goldilock's condition : $\ell_c \ll \ell_w \ll \ell_s$

Semi-classical approx. No wave effects

(e.g.) spectral waveform:

completeness relation

$$\boxed{\hat{1}_{\text{out}} = \int_{p_1',p_2',\ell^n} |p_1', p_2', \ell^n\rangle\langle p_1', p_2', \ell^n|}$$

$$iW(k) = \langle\Psi|\, S^\dagger a(k) S\,|\Psi\rangle$$
$$= \int_{p_1',p_2',\ell^n} \langle\Psi|\, S^\dagger\, |p_1', p_2', \ell^n\rangle \langle p_1', p_2', \ell^n, k|\, S\,|\Psi\rangle$$

We can rewrite the expectation value in terms of scattering amplitudes!

Setup2: Coherent spin formalism

[R. Aoude and A. Ochirov (2021)]

Quantum spin Spin of a massive particle: behaves covariantly under SU(2) group

STATES

Definite spin state $$|s,\{I\}\rangle = \frac{1}{\sqrt{(2s)!}}\,\hat{a}^{\dagger}_{I_1}\cdots\hat{a}^{\dagger}_{I_{2s}}|0\rangle$$

Coherent spin state $$|\alpha\rangle = e^{-\alpha^*_I\alpha^I/2}\sum_{s=0,\frac{1}{2}}^{\infty}\sum_{\{I\}}\frac{\alpha^{I_1}\cdots\alpha^{I_{2s}}}{\sqrt{(2s)!}}|s,\{I\}\rangle$$

SPIN ALGEBRA

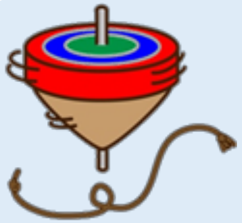
Non-commutative

$$[S^i,S^j] = i\hbar\,\epsilon^{ijk}S^k$$

Classical scaling $$\|\alpha\|^2 = \alpha^*_I\alpha^I \sim \frac{1}{\hbar}$$

=

Classical limit $$\hbar\to 0,\quad s\to\infty,\quad \hbar s = \text{fixed}$$



Classical spin $$\langle S^i\rangle_\alpha \sim \hbar\,\|\alpha\|^2 \sim \mathcal{O}(\hbar^0)$$

Commutative

$$\langle S^iS^j\rangle_\alpha \xrightarrow[\hbar\to 0]{} \langle S^i\rangle_\alpha\,\langle S^j\rangle_\alpha$$

Probing the Classical GR

SCATTERING PROCESS

CONSERVATIVE DYNAMICS

Impulse · *scattering angle* · *EOB Hamiltonian*

$\mathcal{O}(G^2)$: *Damour (2018),…,* $\mathcal{O}(G^5)$: *Bern et al. (2025)*

WAVEFORMS

Memory effect

$\mathcal{O}(G^3)$: *Elkhidir, O'Connell, Sergola, Vázquez-Holm (2023)*

RADIATION & REACTION

radiated energy–momentum · *recoil effect*

$\mathcal{O}(G^3)$ radiative observables:

Herrmann et al. (2021), Damour (2020)

SPIN (Kerr)

Spin multipoles · *Compton*

$\mathcal{O}(G^3 S^4)$: *Akpinar et al. (2025)*

MERGER PROCESS

$O(G)$: tree → ★ $O(G^2)$: 1-loop → $O(G^3)$: 2-loop → $O(G^4)$: 3-loop → $O(G^5)$: 4-loop

[Aoki, Cristofoli, Huang (2024)]

This work!

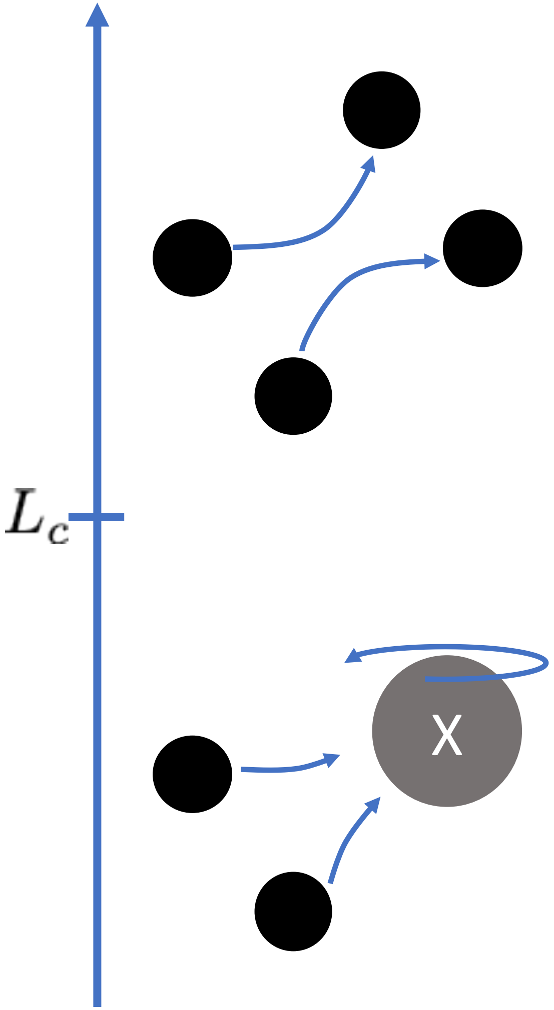
How about Black Hole Mergers ?

Q. How to describe the Black Hole Mergers? [K. Aoki, A. Cristofoli, Y-T. Huang (2024)]

A. Two building blocks. $\xrightarrow{\text{similar ideas}}$

- [K. Aoki, A. Cristofoli, <u>HJ</u>, M. Sergola, K. Yoshimura (2025)]
- [D. Akpinar, K. Aoki, A. Cristofoli, <u>HJ</u>, K. Yoshimura (2026)]

➡ Next Kaho's talk!



① Spectral function

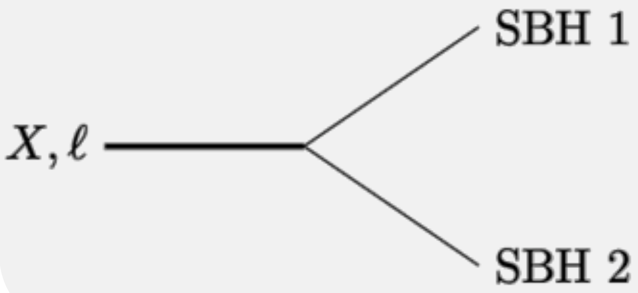
$$\hat{1}_{\mathrm{BH}} = \sum_{\ell,\{I\}} \int \mathrm{d}m_X^2 \int_{p_X} \rho_\ell(m_X^2) \, |p_X, \ell, \{I\}\rangle \langle p_X, \ell, \{I\}|$$

$$\rho_\ell(m_X^2) \propto \theta(L_c - \hbar\ell)$$

describes they merge or not

② Amplitude which describe the Black Hole Mergers

[N. Arkani-Hamed, T. C. Huang, Y. Huang (2018)]



$$: \mathcal{A}^{I_1 \cdots I_{2\ell}}(p_X, \ell | p_1; p_2) = g_\ell \, \langle \boldsymbol{X} | \, p_1 p_2 \, | \boldsymbol{X} \rangle^\ell$$

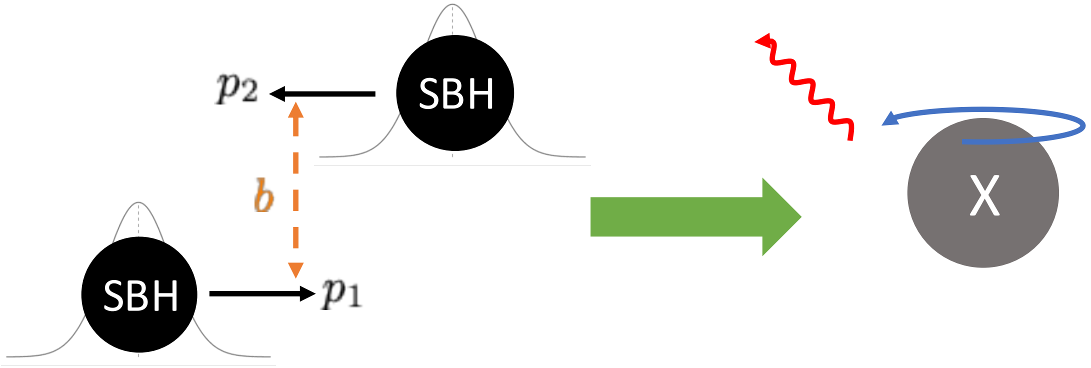
Poincaré symm. determines the 3pt.

Previous work and our aim

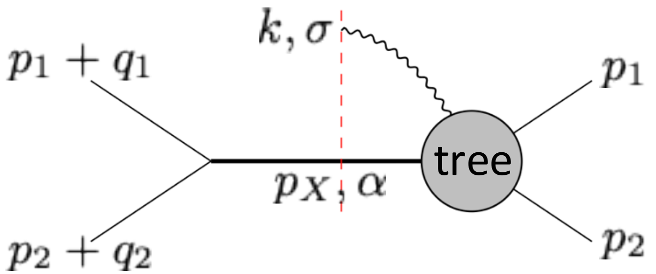
[K. Aoki, A. Cristofoli, Y-T. Huang (2024)]

Gravitational waveform at $\mathcal{O}(G)$

$$h_{\mu\nu} = \kappa \sum_{\sigma} \int_k \varepsilon^{-\sigma}_{\mu\nu} e^{-ik\cdot x} \underbrace{\langle \Psi | \hat{S}^\dagger \hat{a}_\sigma \hat{S} | \Psi \rangle}_{\text{spectral waveform } iW^\sigma(p_i, k, b_i)} + \text{c.c}$$



$$iW^\sigma = \left\langle\!\!\left\langle \int_X \int \prod_{i=1,2} \hat{\mathrm{d}}^4 q_i \hat{\delta}(2p_i \cdot q_i) e^{-ib_i \cdot q_i} \right.\right.$$



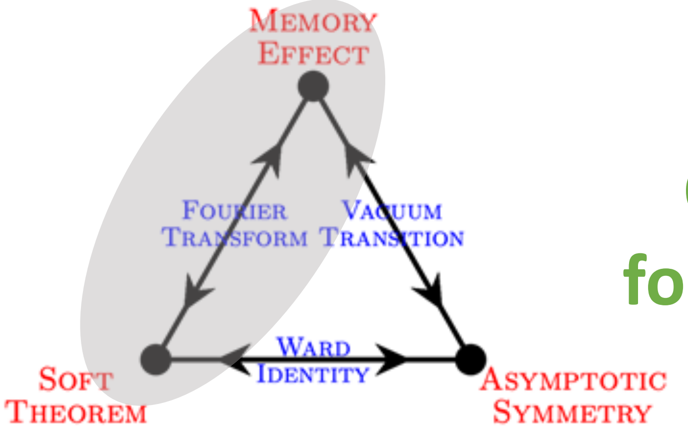
$$\left.\right\rangle\!\!\right\rangle$$

$\kappa = \sqrt{32\pi G}$

$\sigma = \pm$: helicity of graviton

$$= S^{(0)} + S^{(1)} + S^{(2)} + \cdots$$

= **Full Graviton soft operator at $\mathcal{O}(G)$**



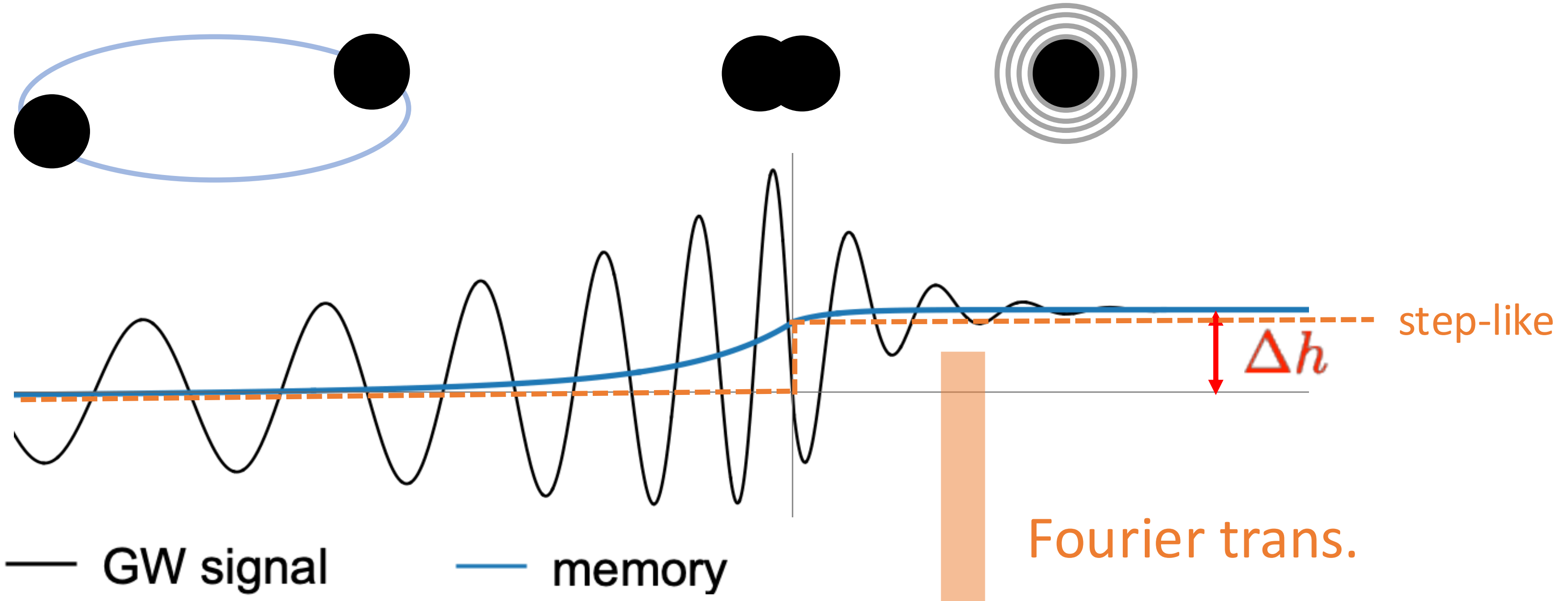
[A. Strominger (2017)]

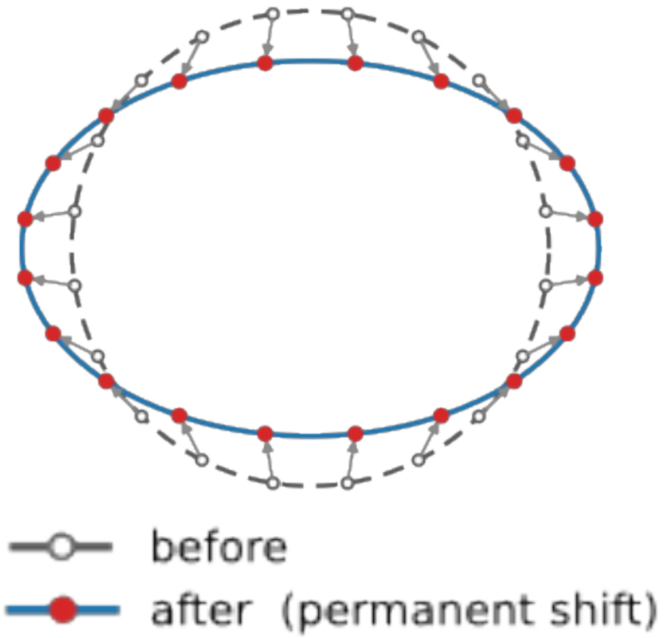
On-shell proof for all spin orders!

Motivation

Interest **Gravitational memory**

$$\Delta h := h(t \to \infty) - h(t \to -\infty)$$





Fourier trans.

Leading soft operator $$S^{(0)} = \frac{\kappa}{2} \sum_{i=1}^{3} \frac{(\epsilon \cdot p_i)^2}{k \cdot p_i} = \mathcal{O}(\omega^{-1})$$ ← Previous work at $\mathcal{O}(G)$!

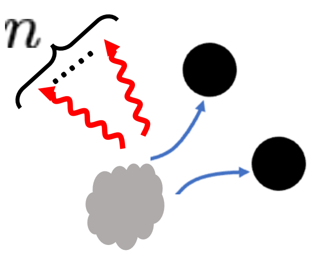
Our aim **Gravitational waveform at $\mathcal{O}(G^2)$** What kind of memory do we have?

Gravitational waveform at $\mathcal{O}(G^2)$ from merger

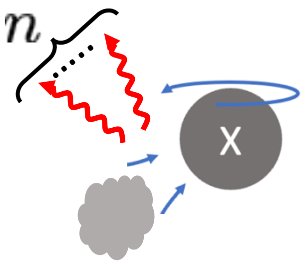
Completeness relation

$$\hat{1}_{\text{out}} = \sum_{n=0}^{\infty} \int_{p_1, p_2, \ell^n} |p_1; p_2; \ell^n\rangle \langle p_1; p_2; \ell^n| + \sum_{n=0}^{\infty} \int_{X, \ell^n} |p_X, \alpha; \ell^n\rangle \langle p_X, \tilde{\alpha}; \ell^n| \,.$$

elastic scattering



inelastic merger



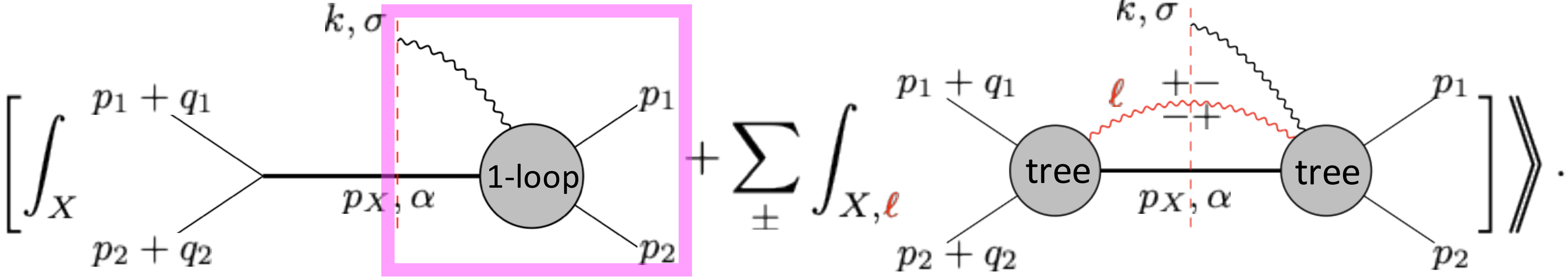
Measure in coherent spin states: $\displaystyle \int_X := \int_{p_X} \int \frac{d^2\alpha\, d^2\tilde{\alpha}}{\pi^2}\, dm_X^2\, \rho_\alpha(m_X^2)$

$$iW_\sigma^{\text{NLO}} = \langle \Psi | \hat{S}^\dagger \hat{a}_\sigma \hat{S} | \Psi \rangle$$

$$= \left\langle\!\!\left\langle \int \prod_{i=1,2} \hat{\mathrm{d}}^4 q_i\, \hat{\delta}(2p_i \cdot q_i)\, e^{-ib_i \cdot q_i} \right.\right.$$

$\sigma = \pm$: helicity of graviton k^μ

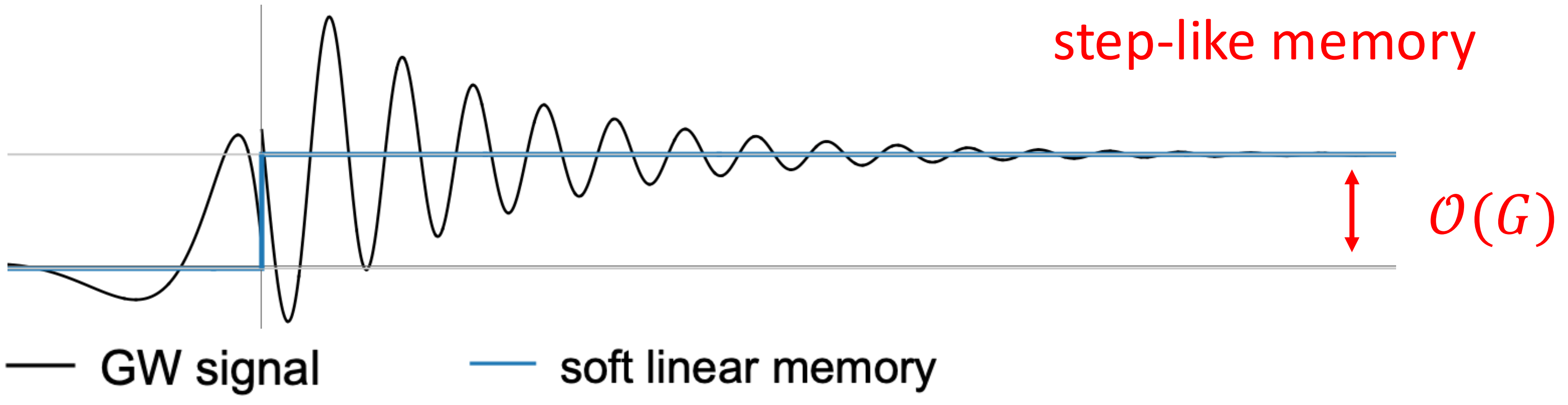
[B. Sahoo, A. Sen (2019), A. Sen (2024)]



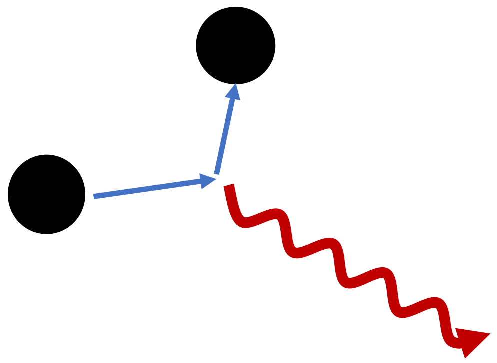
Generalized unitarity & Method of region
FeynGrav & Package-X

Tail memory

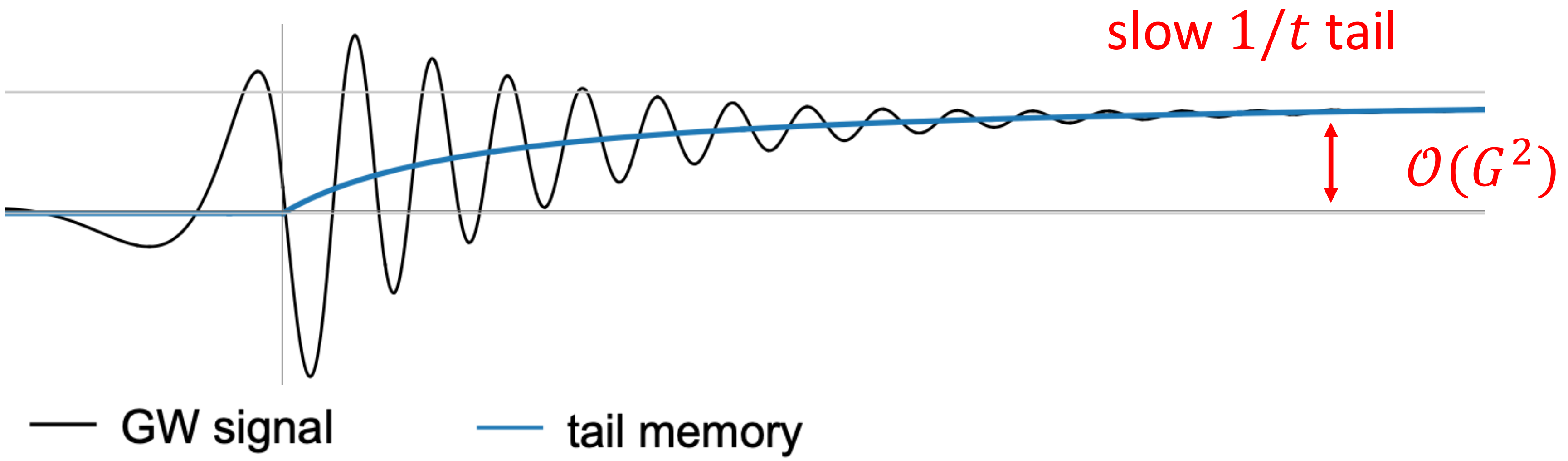
Soft linear memory



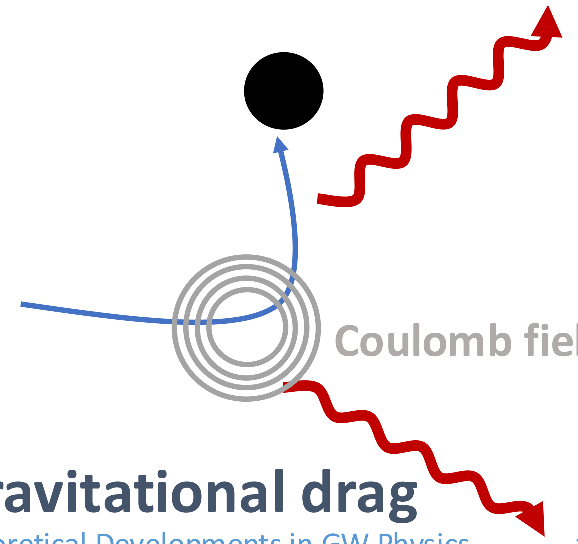
Bremsstrahlung



Tail memory



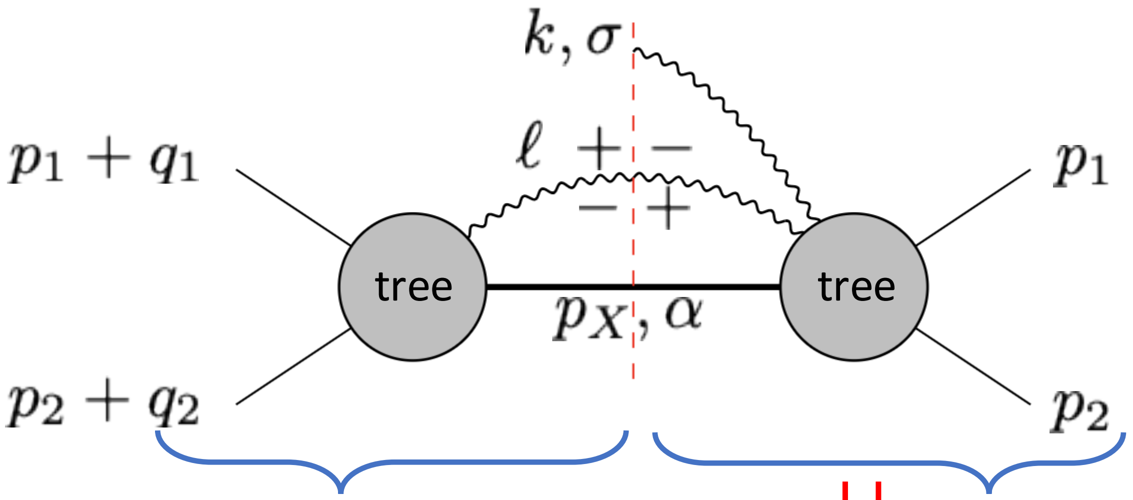
Late-time acceleration



Tree part

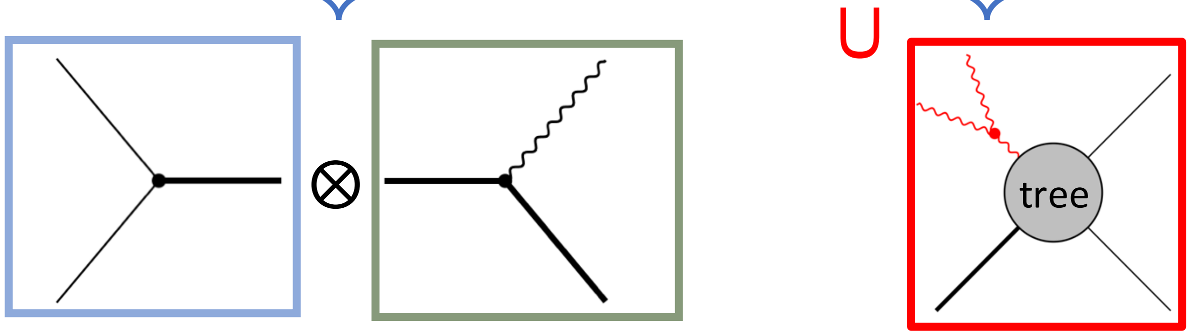
Everything can be computed by local gluing

$$iW_\sigma^{\mathrm{NLO}} = \sum_{\pm} \int_{X,\ell}$$



Exact factorization

∵ coherent spin construction



Momentum conservation **Weinberg's Soft factor** **5pt w/ graviton 3pt vertex**

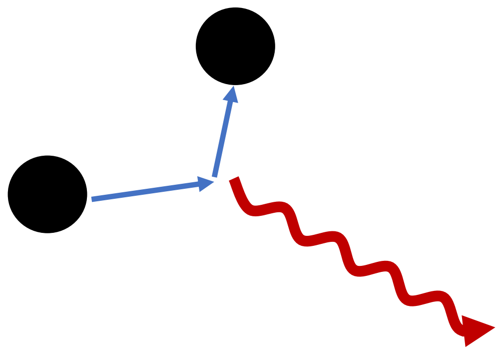
$$= \left(\frac{\kappa}{2}\right)^3 \sum_{i,j=1}^{3} \sum_{\eta=\pm} \int_\ell \frac{[\varepsilon_\sigma(k)\cdot\ell]^2}{k\cdot\ell} \frac{[\varepsilon_\eta(\ell)\cdot p_i]^2}{p_i\cdot\ell} \frac{[\varepsilon_\eta(\ell)\cdot p_j]^2}{p_j\cdot\ell}$$

Non-linear memory!

Non-linear (Christodoulou) memory

[D. Christodoulou (1991), K. S. Thorne (1992)]

Linear Soft memory

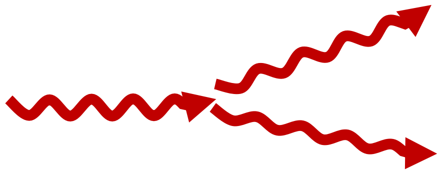


$$\Delta h_{ij}^{\mathrm{TT}} = \frac{4}{r}\Delta \sum_{A=1}^{N} \frac{M_A}{\sqrt{1-v_A^2}} \left(\frac{v_A^i v_A^j}{1 - v_A \cos(\theta_A)} \right)^{\mathrm{TT}}$$

v_A : velocity
M_A : mass
θ_A : angle btw source & detecter

It is sourced by the massive momenta.

Non-linear Soft memory



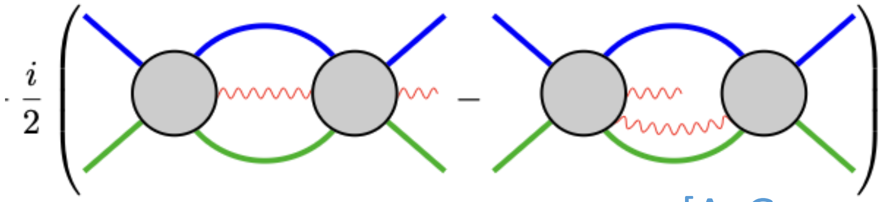
It is sourced by the GW energy flux.

$$\Delta h_{ij}^{\mathrm{TT}} = \frac{4}{r}\int \frac{dE}{d\Omega'} \left(\frac{\xi^{i'} \xi^{j'}}{1 - \cos(\theta')} \right)^{\mathrm{TT}} d\Omega'$$

$d\Omega'$: surrounding the source
ξ' : unit vector from the source & $d\Omega'$
θ' : angle from ξ' to detecter

It appeared at $\mathcal{O}(G^3)$ in $2 \to 2$ scattering case.

But, in our case, it appeared at $\mathcal{O}(G^2)$!



[A. Georgoudis *et al.* (2025)]

Summary and outlook

Agenda **Gravitational waveform from merger at $\mathcal{O}(G^2)$**

Q. What kind of memory do we have?

A. **Tail memory** and **non-linear soft memory**

NEW!

Outlook

- Generalization to **KBH** merger.
- **Higher loops** and its connection to **Soft theorem** and **BMS symmetry**.
- Investigating the **re-summation structure**.

  (e.g.) Sommerfeld enhancement?
- Connection with **inspiral** and **ringdown**.
- **Double Copy** for the Black Hole Mergers?

Thank you !